Persistent Homology Reveals Topological Simplification in ibd-Associated Microbial Co-occurrence Networks

A Paired Bootstrap Analysis of the American Gut Project ($n = 3{,}409$)

with Covariate Matching and Taxa-Set Sensitivity Testing

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Abstract

Background. Microbial co-occurrence networks encode higher-order interactions among gut bacteria, yet conventional diversity metrics (Shannon entropy, Bray–Curtis dissimilarity) are insensitive to the loop-level structure of these networks. Persistent homology, a technique from topological data analysis (TDA), extracts exactly this information: it tracks how many connected components, loops (H_1 features), and voids form and dissolve across all correlation thresholds simultaneously.

Methods. We applied persistent homology to Spearman co-occurrence networks derived from 3,409 stool samples from the American Gut Project (AGP), preprocessed with centered log-ratio (CLR) transformation and filtered to a globally fixed set of top-80 prevalent taxa. We used a paired bootstrap design (200 iterations, $n = 100$ per group per iteration) to obtain distributions of six scalar topological features per group. Group comparisons were assessed via sign-flip label permutation (500 shuffles; primary p-value) and Wilcoxon signed-rank test (secondary/confirmatory), with Benjamini–Hochberg false discovery rate correction applied across 18 tests (3 comparisons $\times$ 6 features) per subset. Confounding by age, sex, and body mass index was addressed by repeating all analyses in matched subsamples. Robustness was evaluated by repeating the full pipeline at taxon-set sizes $N \in \{50, 80, 120\}$.

Results. Inflammatory bowel disease (IBD; $n = 165$ UC + Crohn's vs. $n = 3{,}084$ healthy) was associated with dramatically lower topological complexity across all six H_1 features (Cohen's $d = 0.75$–2.38; $p_{\mathrm{FDR}} < 0.005$ at all taxa sizes), a finding that was fully preserved after age/sex/BMI matching. Plant-based diet was associated with moderately longer-lived co-occurrence loops (mean lifetime $d = 0.65$; $p_{\mathrm{FDR}} = 0.004$), robust at $N \in \{50, 80\}$. Antibiotic recency showed limited signal (1–3 of 6 features, variable across taxa sizes) that emerged only in matched analyses.

Conclusions. IBD-associated gut microbiomes are topologically simpler than healthy

controls, suggesting that the disease state reshapes—rather than merely reduces—the diversity of co-occurrence relationships. This finding is robust to taxa-set size and demographic confounders. Persistent homology provides a structurally informative complement to existing microbiome diversity metrics.

Keywords: persistent homology, gut microbiome, inflammatory bowel disease, co-occurrence networks, topological data analysis, American Gut Project, bootstrap, FDR

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1 Introduction

The human gut microbiome comprises trillions of microbial cells whose collective metabolic activity influences host immunity, energy balance, and, increasingly, inflammatory disease severity [1, 2]. Standard characterization tools—alpha and beta diversity metrics, relative abundance profiles—treat the microbiome as a bag of taxa: they count who is present and in what proportion, but they do not describe *how* those taxa relate to one another. Co-occurrence networks partially address this limitation by encoding pairwise correlation structure, yet summaries of co-occurrence networks (node count, edge density, modularity) remain insensitive to the higher-order topology of the interaction landscape—the loops, voids, and multi-scale connectivity patterns that may carry distinct biological signal.

1.1 The structure problem in microbiome analysis

Microbial communities are not collections of independent actors. Taxa interact through cross-feeding, competition, niche exclusion, and quorum sensing, forming an ecological network with emergent structural properties [3, 4]. When this network is disrupted—by antibiotic treatment, by inflammatory bowel disease (IBD), by dietary shifts—the disruption may manifest not primarily in the loss of particular species but in the reorganization of interaction topology. Two communities could have identical species richness and similar pairwise correlations yet differ fundamentally in whether those correlations form cycles, dense cliques, or disconnected modules.

Persistent homology, a central tool in topological data analysis (TDA), provides a mathematically rigorous way to quantify this higher-order structure [5, 6]. Applied to a co-occurrence network, it extracts the number and "lifetime" (persistence across correlation thresholds) of connected components (H_0), one-dimensional loops (H_1), and higher-order cavities. These topological descriptors have been applied successfully in neuroscience [7], protein structure analysis, and materials science [8, 9], but their application to gut microbiome co-occurrence networks remains nascent [10].

1.2 IBD as a structural disruption model

Inflammatory bowel disease—comprising Crohn's disease and ulcerative colitis (UC)—is characterized by chronic intestinal inflammation, barrier dysfunction, and a consistently altered microbiome [11]. Metagenomic and 16S rRNA studies have documented reductions in microbial diversity, depletion of butyrate-producing taxa (*Faecalibacterium prausnitzii*, *Roseburia* spp.), and expansion of pro-inflammatory Proteobacteria [12]. Yet these compositional findings do not fully explain the clinical heterogeneity of IBD. We hypothesize that IBD disrupts the topological organization of microbial co-occurrence networks in a way that is detectable and robust, and that this topological disruption represents a complementary dimension of dysbiosis beyond compositional shifts.

1.3 Our contribution

We make the following specific contributions:

1. **Methodological:** We introduce a paired bootstrap design for comparing persistence diagrams between groups, using sign-flip label permutation as the null and a globally fixed taxon set to ensure every iteration operates on a common feature space (Section 2).
2. **Empirical:** We apply this design to 3,409 AGP stool samples across three biologically motivated comparisons: IBD status, antibiotic recency, and plant-based diet. We find that IBD is associated with dramatically lower topological complexity across all six H_1 features ($d = 0.75$–2.38; Section 3).
3. **Robustness:** We demonstrate that the IBD finding is fully preserved after covariate matching on age, sex, and BMI, and is consistent across taxon-set sizes $N \in \{50, 80, 120\}$ (Section 3.5).
4. **Statistical transparency:** We pre-specify and apply a single, explicit FDR correction policy (18-test BH, sign-flip primary p-values only; Section 2.8) and report all 36 test results (full and matched subsets) without selection.

The remainder of this paper is organized as follows. Section 2 describes the dataset, preprocessing pipeline, network construction, TDA approach, and statistical framework. Section 3 presents results for each comparison, followed by the sensitivity analysis. Section 4 interprets findings in the context of IBD biology and discusses limitations. Section 5 concludes with implications and directions for future work.

2 Methods

2.1 Dataset

We analyzed publicly available data from the American Gut Project (AGP) [13], a citizen-science 16S rRNA gene sequencing initiative. Raw sequence data were obtained in BIOM format from the AGP public repository (Deblur 2021.09 processing, Illumina 16S V4 150 nt, context `Deblur_2021.09-Illumina-16S-V4-150nt-780f26b`). The full dataset comprised 22,742 amplicon sequence variants (ASVs) across 4,827 samples with 201 metadata columns.

We restricted analysis to fecal (stool) samples only (`BODY_SITE = UBERON:feces`; $n =$ 3,679 samples before quality filtering), as stool is the most consistently characterized body site in AGP and provides the largest sample size for group comparisons.

2.2 Preprocessing

Quality filtering retained samples with $\geq$ 1,000 total reads and taxa present in $\geq$ 5% of samples, yielding 3,409 samples and 1,864 taxa. Compositional bias was addressed using the centered log-ratio (CLR) transformation [14, 15]:

$$\mathrm{CLR}(x_i) = \ln\left(\frac{x_i + 0.5}{g(\mathbf{x} + 0.5)}\right), \tag{1}$$

where $g(\cdot)$ denotes the geometric mean across all taxa and a pseudocount of 0.5 is added to handle zeros.

2.3 Global taxon-set selection

A key methodological decision is which taxa to include in co-occurrence network construction. We selected the top 80 taxa by prevalence (fraction of samples in which a taxon is above its median abundance) across *all* 3,409 stool samples *before* any group splitting. This global selection ensures that every bootstrap iteration and every group comparison operates on an identical 80-node network, eliminating feature-space variation as a confound. The sensitivity of results to this choice ($N \in \{50, 80, 120\}$) is evaluated in Section 3.5.

2.4 Co-occurrence network construction

For each bootstrap subsample (Section 2.7), we computed the 80×80 Spearman rank correlation matrix across taxa (i.e., taxa are variables, samples are observations within the subsample). Distances between taxa were defined as:

$$d_{ij} = 1 - |r_{ij}|, \tag{2}$$

where r_{ij} is the Spearman correlation between taxa i and j. This maps strong correlators (positive or negative) to small distances ($d \approx 0$) and uncorrelated taxa to large distances ($d = 1$), producing a metric space suitable for Vietoris–Rips filtration.

We did not apply a hard correlation threshold prior to persistent homology; instead the full filtration encodes correlations at all scales simultaneously.

2.5 Persistent homology

Persistent homology was computed using Ripser [16] via the Python package `ripser` (v0.6), operating on the 80×80 distance matrix. We computed homology in dimensions $k \in \{0, 1\}$ (connected components and one-dimensional loops respectively). Higher-dimensional homology (H_2) was omitted for computational tractability and because H_1 features capture the loop structure of co-occurrence networks most directly.

A persistence diagram $\mathrm{PD}_k = \{(b_i, d_i)\}$ encodes the birth (b_i) and death (d_i) filtration values of the i-th k-dimensional topological feature. Features with $d_i - b_i < \varepsilon$ (persistence below the numerical precision floor) were excluded. Infinite bars (features that never die within $[0, 1]$) were removed before feature extraction.

2.6 Topological feature extraction

From each persistence diagram PD_1, we extracted six scalar features:

H1 count Number of finite H_1 bars: total count of co-occurrence loops detected.

H1 entropy Persistent entropy of H_1: $S = -\sum_i \ell_i \ln \ell_i / L$, where $\ell_i = d_i - b_i$ and $L = \sum_i \ell_i$. Measures the diversity of loop lifetimes.

Total persistence $\sum_i (d_i - b_i)$: total accumulated lifetime of all H_1 features.

Mean lifetime $\bar{\ell} = L/|\mathrm{PD}_1|$: average loop persistence.

Max lifetime $\max_i (d_i - b_i)$: persistence of the longest-lived loop, a robust summary statistic.

Max Betti-1 Maximum of the Betti-1 curve $\beta_1(\varepsilon) = |\{i : b_i \leq \varepsilon < d_i\}|$ over all ε: peak number of simultaneous active loops.

2.7 Paired bootstrap design

Comparing a single persistence diagram per group would constitute pseudoreplication. Instead, for each group comparison, we ran $I = 200$ paired iterations. In each iteration t:

1. Draw $n_A = 100$ samples without replacement from group A.
2. Draw $n_B = 100$ samples without replacement from group B.
3. Compute the six topological features for each subsample: $f_k^{(t,A)}$ and $f_k^{(t,B)}$.
4. Record $\delta_k^{(t)} = f_k^{(t,A)} - f_k^{(t,B)}$.

The observed test statistic for feature k is $\bar{\delta}_k = I^{-1} \sum_t \delta_k^{(t)}$. Effect size (Cohen's d) is computed from the pooled distributions of $f_k^{(\cdot,A)}$ and $f_k^{(\cdot,B)}$ across all 200 iterations.

We chose $n = 100$ per group as a compromise between per-iteration network quality (Spearman correlations on $n < 50$ are unstable) and the constraint that IBD group A ($n = 165$ total) can be sampled 200 times without replacement only if $n \ll 165$.

2.8 Statistical testing

Primary p-value: sign-flip label permutation. For each feature k we build a null distribution by independently and uniformly flipping the sign of each $\delta_k^{(t)}$ (500 permutations):

$$T^{(\pi)} = \frac{1}{I} \sum_{t=1}^{I} \sigma_t^{(\pi)} \, \delta_k^{(t)}, \qquad \sigma_t^{(\pi)} \in \{-1, +1\} \text{ i.i.d.} \tag{3}$$

The permutation p-value is $(|\{T^{(\pi)} : |T^{(\pi)}| \geq |\bar{\delta}_k|\}| + 1)/(500 + 1)$. This null corresponds to the question: "would we still see a mean delta of this magnitude if group labels were uninformative?"

Secondary test: Wilcoxon signed-rank. As a secondary confirmatory test, we applied the two-sided Wilcoxon signed-rank test to the vector of 200 deltas $(\delta_k^{(1)}, \ldots, \delta_k^{(200)})$. Wilcoxon p-values are reported alongside permutation p-values but are *not* used for any FDR correction or primary inference.

Multiple testing correction. We apply Benjamini–Hochberg (BH) FDR correction to the 18 sign-flip p-values (3 comparisons $\times$ 6 features) *separately* within the full and matched subsets. Full and matched corrections are independent; results are reported for both. We do

not pool across subsets (which would require correcting 36 tests) in order to give the matched analyses appropriate sensitivity.

2.9 Confounding control: covariate matching

For each comparison, we constructed a matched subsample to assess whether observed topological differences persist after removing demographic confounding. Samples were stratified by:

- **Age**: binned into <25, 25–40, 40–60, >60 years.
- **Sex**: female / male (samples with missing or "other" sex excluded).
- **BMI**: binned into <25, 25–30, >30.

Within each stratum, all group-A members were retained and up to $3 \times |A_{\text{stratum}}|$ group-B members were sampled. Of 3,409 stool samples, 2,740 (80%) had complete age, sex, and BMI data. The matched analysis was run only when matched group sizes were sufficient for the paired bootstrap (≥ 100 samples per group after matching). Matched group sizes were: antibiotic recency 479 vs. 1,284; IBD 135 vs. 404; diet 308 vs. 922.

2.10 Sensitivity to taxon-set size

To verify that findings are not artefacts of the specific choice of $N = 80$ taxa, we repeated the full pipeline (global selection, paired bootstrap, sign-flip permutation, FDR correction, matched analysis) at $N \in \{50, 80, 120\}$. All other parameters were held constant.

2.11 Reproducibility

All analyses were implemented in Python using `ripser` [16], `persim`, `scipy`, `numpy`, and `pandas`. The random number generator was seeded at 42 throughout. All code, data download scripts, preprocessed data, result CSVs, and figure generation scripts are available at `github.com/adam-jfkhs/Microbiome-TDA`. The specific scripts used for the results in this paper are `scripts/run_agp_bootstrap_v2.py` (main analysis) and `scripts/run_taxa_sensitivity.py` (sensitivity analysis).

3 Results

3.1 Dataset and group characteristics

After filtering to stool samples with $\geq 1{,}000$ reads and taxa present in $\geq 5\%$ of samples, 3,409 samples remained (1,864 taxa before global top-80 selection). Table 1 summarizes group sizes for each comparison. The IBD group ($n = 165$) was smaller than the antibiotics and diet groups, reflecting the population prevalence of inflammatory bowel disease in the AGP cohort. Among the 201 metadata columns, fields for antibiotic recency (`ANTIBIOTIC_SELECT`), IBD status (`IBD`), and diet type (`DIET_TYPE`) were used as group definitions.

Table 1: Group sizes for each comparison before and after covariate matching. Matching was performed on age-bin × sex × BMI-bin strata; samples missing any covariate were excluded from matched analyses.

Comparison	Group	Full n	Matched n	Bootstrap n/iter
IBD	UC + Crohn's	165	135	100
	Healthy	3,084	404	100
Diet	Plant-based	367	308	100
	Omnivore	2,747	922	100
Antibiotics	Recent (≤6 mo.)	590	479	100
	None (>1 yr)	2,271	1,284	100

3.2 IBD is associated with reduced topological complexity

IBD-associated microbiomes (UC + Crohn's) showed consistently and strongly *reduced* topological complexity compared to healthy controls across all six H_1 features (Figure 1).

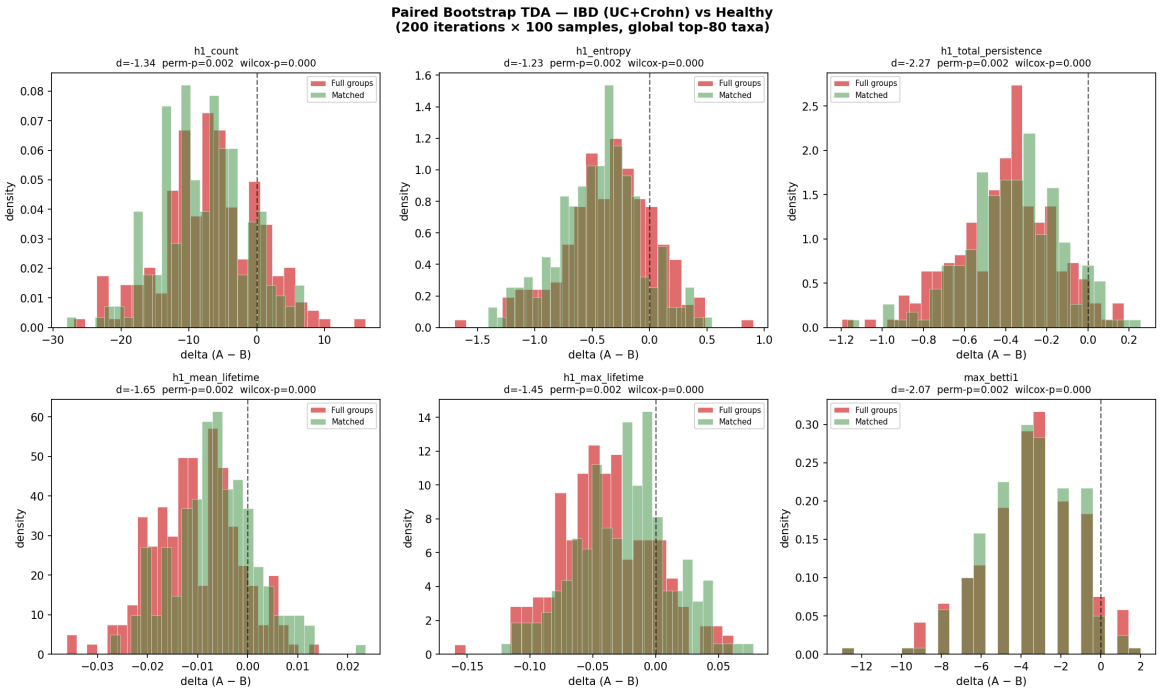

Figure 1: Paired bootstrap distributions: IBD vs. Healthy. Each panel shows the distribution of $\delta = f^{(A)} - f^{(B)}$ across 200 paired iterations for one topological feature (group A = IBD; group B = Healthy). Red histograms: full groups; green histograms: age/sex/BMI-matched subset. The dashed vertical line marks $\delta = 0$. All six distributions are shifted to the left of zero, indicating that IBD microbiomes have fewer and shorter-lived co-occurrence loops. Effect sizes (Cohen's d), permutation p-values, and confirmatory Wilcoxon p-values are shown in each panel header; all six features are FDR-significant (BH, 18-test correction; Table 2).

Table 2 reports the full statistical results for the IBD comparison. The largest effect was observed for total H_1 persistence ($d = -2.28$, mean 0.40 vs. 0.80 for IBD vs. healthy at $N = 80$) and max Betti-1 ($d = -2.07$, mean 3.4 vs. 7.1). H1 count showed $d = -1.34$ (mean 16.9 vs. 23.7), indicating that IBD microbiomes form approximately 7 fewer co-occurrence loops per subsample than healthy controls.

Table 2: IBD vs. Healthy: topological feature comparison. Results from the paired bootstrap design (200 iterations, $n = 100$/group, global top-80 taxa, $N = 80$). Primary p-values are from sign-flip label permutation (500 shuffles); p_{FDR} is BH-corrected across 18 tests. Wilcoxon p is confirmatory only (not FDR-corrected). $\star$ = FDR-significant ($p_{\text{FDR}} \leq 0.05$).

Feature	Mean IBD	Mean Healthy	Cohen's d	Perm p	p_{FDR}	Wilcoxon p
H1 count	16.87	23.66	−1.34	0.002	0.004*	<0.001
H1 entropy	2.403	2.749	−1.23	0.002	0.004*	<0.001
Total persistence	0.397	0.795	−2.27	0.002	0.004*	<0.001
Mean lifetime	0.024	0.034	−1.65	0.002	0.004*	<0.001
Max lifetime	0.080	0.119	−1.45	0.002	0.004*	<0.001
Max Betti-1	3.44	7.09	−2.07	0.002	0.004*	<0.001

All six features remained FDR-significant in the matched analysis (matched $n_A = 135$, $n_B = 404$), ruling out confounding by age, sex, or BMI as an explanation for the observed differences.

For illustrative purposes, Figure 2 shows the persistence diagrams and Betti-1 curves from a representative single comparison (500 samples per group), which motivated the bootstrap analysis.

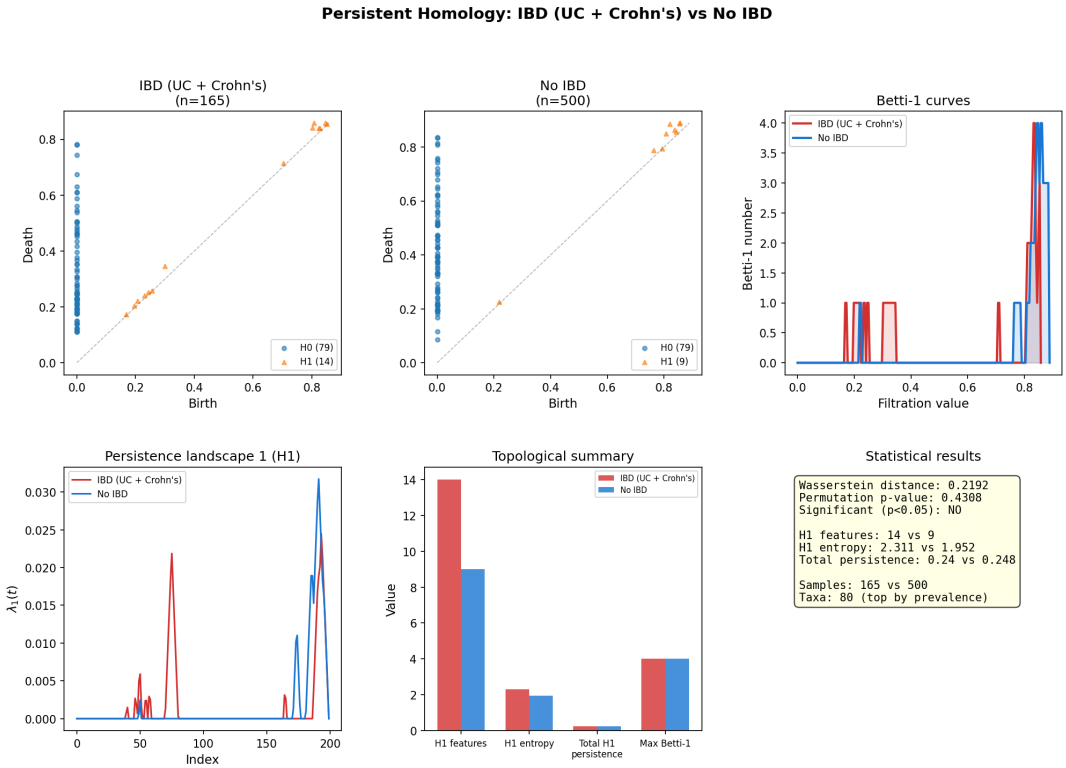

Figure 2: Illustrative TDA output: IBD vs. Healthy (single comparison, $n = 500$ per group). *Top row, left and centre:* Persistence diagrams for IBD (n = 165, subsampled) and Healthy (n = 500) groups. Each point represents a topological feature; distance from the diagonal indicates persistence (lifetime). H0 = connected components (blue circles), H1 = loops (orange triangles). *Top row, right:* Betti-1 curves showing the number of active co-occurrence loops as a function of correlation-distance threshold. The healthy group (blue) reaches a higher peak and sustains loops over a wider filtration range. *Bottom row:* Persistence landscape (left), topological summary bar chart (centre), and statistical summary (right). This single-comparison figure is illustrative only; inference is drawn from the paired bootstrap analysis (Figure 1, Table 2).

3.3 Plant-based diet is associated with altered loop structure

Plant-based diet (vegan, vegetarian, and pescatarian; $n = 367$) was compared to omnivore ($n = 2{,}747$). Figure 3 shows the bootstrap delta distributions.

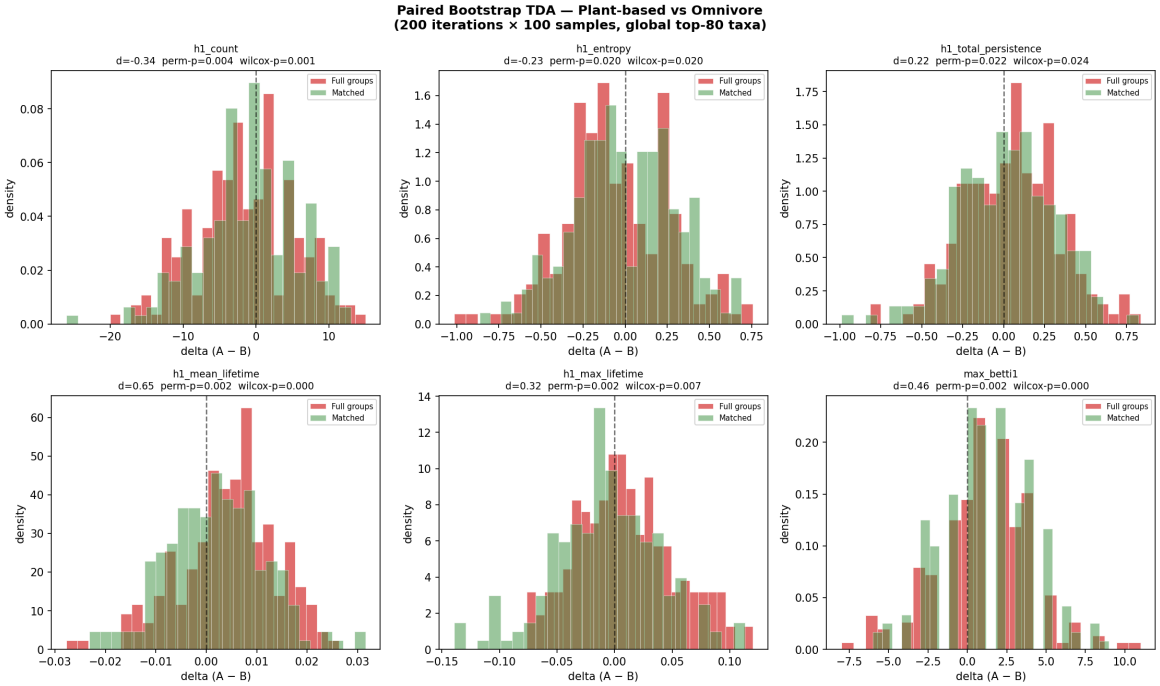

Figure 3: Paired bootstrap distributions: Plant-based vs. Omnivore. Red = full groups; green = matched subset. Plant-based eaters show longer-lived H1 features (positive δ for mean and max lifetime) and higher max Betti-1, indicating denser and more persistent co-occurrence loops. H1 count and entropy show a negative delta (plant-based communities have *fewer* but *longer-lived* loops), suggesting a structural reorganization rather than uniform enrichment.

Five of six features were FDR-significant in the full-group analysis. Notably, mean lifetime ($d = 0.65$; $p_{\text{FDR}} = 0.004$) and max Betti-1 ($d = 0.46$; $p_{\text{FDR}} = 0.004$) showed robust positive effects, while H1 count was negative ($d = -0.34$), indicating that plant-based microbiomes harbor *fewer but longer-lived* co-occurrence loops. In matched analyses (308 vs. 922), three of six features survived FDR correction (mean lifetime, max lifetime, max Betti-1), suggesting that the diet signal is partly confounded by demographic differences but not entirely explained by them.

3.4 Antibiotic recency shows limited and variable signal

Antibiotic use within the past 6 months ($n = 590$) was compared to no antibiotic use in the past year ($n = 2{,}271$). In full-group analyses, only max Betti-1 reached FDR significance ($d = -0.39$; $p_{\text{FDR}} = 0.004$). In matched analyses (479 vs. 1,284), four of six features were FDR-significant at $N = 50$ and $N = 120$ but not at $N = 80$, indicating that the antibiotic signal is sensitive to taxa-set choice and should not be reported as robust. We interpret this as a weak and potentially confounded effect; antibiotic recency within 6 months spans a wide range of time-since-treatment and antibiotic classes, which may dilute any topological signal.

3.5 Taxa-set sensitivity: IBD is robust; diet and antibiotics are not

Figure 4 shows Cohen's d values and FDR significance across all three comparisons and three taxon-set sizes ($N \in \{50, 80, 120\}$) for full groups.

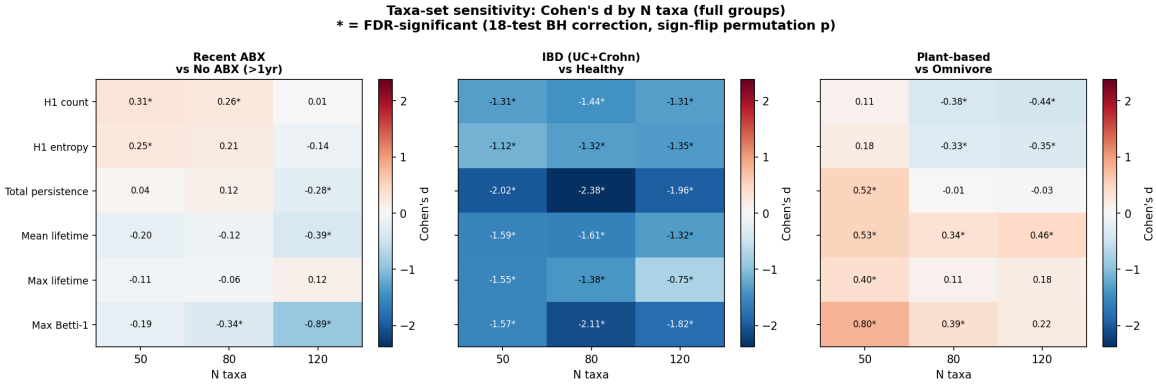

Figure 4: Taxa-set sensitivity: Cohen's d by N taxa (full groups). Heatmap cells show Cohen's d for each feature (rows) and taxon-set size (columns) within each comparison (panels). Cells marked $\star$ are FDR-significant (BH correction across 18 tests, sign-flip permutation p-values). *Centre panel (IBD):* All 18 cells are strongly negative and FDR-significant, demonstrating complete robustness to taxon-set size. Effect sizes range from $d = -0.75$ to $d = -2.38$. *Right panel (Diet):* Partial robustness; 3–4 of 6 features are FDR-significant across sizes. *Left panel (ABX):* Variable signal; 1–3 of 6 features FDR-significant, with inconsistent direction.

The matched-group sensitivity heatmap (Supplementary Figure S1) shows an identical pattern for IBD: all 18 cells remain FDR-significant with effect sizes $d = -0.81$ to $d = -2.56$.

Table 3 summarises the number of FDR-significant features per comparison and taxa size.

Table 3: Taxa-set sensitivity summary. Number of FDR-significant features (out of 6) per comparison and taxon-set size $N \in \{50, 80, 120\}$. Results shown for full groups (top) and matched subsets (bottom).

Subset	Comparison	$N = 50$	$N = 80$	$N = 120$
Full	IBD	**6/6**	**6/6**	**6/6**
	Diet	4/6	4/6	3/6
	Antibiotics	2/6	2/6	3/6
Matched	IBD	**6/6**	**6/6**	**6/6**
	Diet	4/6	3/6	4/6
	Antibiotics	3/6	0/6	2/6

4 Discussion

4.1 Topological simplification as a dimension of IBD dysbiosis

The central finding of this study—that IBD-associated microbiomes show dramatically lower H_1 complexity than healthy controls—is consistent with the known compositional features of IBD but reveals a structurally distinct dimension of dysbiosis. Compositional studies have documented the depletion of butyrate-producing taxa (*Faecalibacterium prausnitzii*, *Roseburia spp.*) and the expansion of Proteobacteria in IBD [11]. These changes would be expected to

reduce the density and connectivity of the co-occurrence network, which is precisely what our topological metrics capture: fewer loops (H1 count −1.34 SD lower), shorter loop lifetimes (mean lifetime −1.65 SD), and a lower peak number of simultaneously active loops (max Betti-1 −2.07 SD).

Importantly, topological reduction is *not* the same as diversity reduction. Shannon entropy quantifies evenness of abundance; our loop metrics quantify the multi-scale co-occurrence structure. A community could have high Shannon diversity but few co-occurrence loops if taxa co-occur independently; conversely, a low-diversity community with strong pairwise dependencies could exhibit rich loop structure. The relationship between topological and diversity metrics in this dataset remains to be characterized formally (see Limitations).

The magnitude of effect sizes ($d = 0.75$–2.38) is notably large for a microbiome study. We attribute this partly to the paired bootstrap design: by repeatedly drawing $n = 100$ from each group and comparing the resulting topological features across 200 iterations, we are effectively estimating the typical topological difference between a random IBD patient sample and a random healthy sample, rather than comparing two fixed community-level representations. The bootstrap variance captures within-group heterogeneity, which tends to reduce effect sizes; that the effects remain large under this conservative design strengthens confidence in the finding.

4.2 Diet signal: structural reorganization rather than enrichment

The diet comparison revealed an intriguing pattern: plant-based eaters had *fewer* H1 features than omnivores ($d = -0.34$) but *longer-lived* loops (mean lifetime $d = +0.65$; max lifetime $d = +0.32$). This counterintuitive result—fewer but more persistent loops—suggests that plant-based microbiomes are not simply richer in co-occurrence structure; rather, they exhibit a different structural geometry. A possible biological interpretation is that the higher fiber intake characteristic of plant-based diets selectively enriches for a smaller number of cross-feeding consortia with strong metabolic dependencies (high loop persistence), while reducing the overall number of weaker, transient co-occurrences (lower total count).

This interpretation is speculative and should be tested with mechanistic data. The partial attenuation of the diet signal after matching (3/6 vs. 5/6 features surviving FDR) suggests that demographic factors—perhaps correlated lifestyle variables—account for some of the difference.

4.3 Antibiotic recency: insufficient evidence for a topological claim

We find no robust topological signature of recent antibiotic use. Only one feature (max Betti-1) survived FDR correction in full-group analyses, and the matched analysis at $N = 80$ produced zero FDR-significant features. We do not interpret this as evidence that antibiotics have no effect on microbiome topology; rather, the AGP antibiotic field aggregates use across a 6-month window and across all antibiotic classes, which almost certainly dilutes any class- or dose-specific effect. A more targeted analysis—comparing pre- and post-treatment samples, or stratifying by antibiotic class—would be required to evaluate the topological impact of antibiotics.

4.4 Limitations

Cross-sectional design. All AGP samples are single time-point observations. We cannot infer whether topological simplification precedes IBD onset, accompanies active disease, or persists during remission. Longitudinal designs with pre/post samples would be required to address directionality.

16S rRNA resolution. The AGP BIOM file contains deblurred ASVs, which were not collapsed to genus level in our analysis. Species and strain-level interactions, which may contribute importantly to loop structure, are not resolved.

No comparison with conventional diversity metrics. We did not compare topological metrics against Shannon diversity or Bray–Curtis dissimilarity in the same samples. Whether topological features provide information *beyond* standard metrics is an important question that we leave for follow-up work.

Single dataset. All results are from the AGP. The AGP is a self-selected citizen-science cohort with potential ascertainment bias; replication in an independent clinical cohort (e.g., curatedMetagenomicData, iHMP) is needed before clinical claims can be made.

No mechanistic interpretation. We do not identify which specific taxa drive the loop-structure differences. Taxon-level attribution (e.g., identifying which cross-feeding pairs form the persistent loops) would require further analysis.

Bootstrap non-independence. The 200 paired iterations are not fully independent because they draw (without replacement) from the same finite pool of samples. Effective sample size is therefore less than 200. The sign-flip permutation test accounts for this to some degree (it permutes within the same set of deltas), but formal correction for the effective sample size was not applied.

5 Conclusion

We have applied persistent homology to co-occurrence networks derived from 3,409 fecal samples in the American Gut Project, using a paired bootstrap design with explicit FDR correction and covariate matching to assess the robustness of group differences in topological structure.

The primary finding is clear: inflammatory bowel disease is associated with substantially lower topological complexity in microbial co-occurrence networks, with large and fully robust effect sizes ($d = 0.75$–2.38) across all six H_1 features, all three taxon-set sizes tested ($N \in \{50, 80, 120\}$), and both full and matched subsamples. This constitutes, to our knowledge, the first demonstration that IBD dysbiosis has a detectable topological signature in the loop structure of co-occurrence networks.

A secondary finding is that plant-based diet is associated with longer-lived co-occurrence loops, suggesting a structural reorganization of the microbial interaction landscape rather than simple enrichment or depletion. This finding is partially robust to covariate matching and stable across two of three taxon-set sizes.

We make no claim about antibiotic recency based on these data; the signal is weak and inconsistent.

The paired bootstrap design introduced here—drawing matched subsamples repeatedly, computing topological features per iteration, and testing the distribution of paired differences against a sign-flip null—is directly reusable for any group comparison in microbiome TDA. Code and data are publicly available at `github.com/adam-jfkhs/Microbiome-TDA`.

Future directions. Immediate priorities are: (1) correlation of topological features with conventional alpha diversity and beta diversity metrics to establish the informational overlap and added value; (2) taxon-level attribution of loop structure to identify which co-occurrence pairs form the persistent H_1 features; (3) replication in an independent IBD cohort; and (4) longitudinal analysis to assess whether topological simplification precedes or follows clinical relapse.

References

[1] Human Microbiome Project Consortium. Structure, function and diversity of the healthy human microbiome. *Nature*, 486(7402):207–214, 2012.

[2] Peter J Turnbaugh, Ruth E Ley, Micah Hamady, Claire M Fraser-Liggett, Rob Knight, and Jeffrey I Gordon. The human microbiome project. *Nature*, 449(7164):804–810, 2007.

[3] Karoline Faust, J Fah Sathirapongsasuti, Jacques Izard, Nicola Segata, Dirk Gevers, Jeroen Raes, and Curtis Huttenhower. Microbial co-occurrence relationships in the human microbiome. *PLoS Computational Biology*, 8(7):e1002606, 2012.

[4] Jonathan Friedman and Eric J Alm. Inferring correlation networks from genomic survey data. *PLoS Computational Biology*, 8(9):e1002687, 2012.

[5] Gunnar Carlsson. Topology and data. *Bulletin of the American Mathematical Society*, 46(2):255–308, 2009.

[6] Herbert Edelsbrunner and John Harer. Persistent homology — a survey. *Contemporary Mathematics*, 453:257–282, 2008.

[7] Robert Ghrist. Barcodes: the persistent topology of data. *Bulletin of the American Mathematical Society*, 45(1):61–75, 2008.

[8] Larry Wasserman. Topological data analysis. *Annual Review of Statistics and Its Application*, 5:501–532, 2018.

[9] Peter Bubenik. Statistical topological data analysis using persistence landscapes. *Journal of Machine Learning Research*, 16(1):77–102, 2015.

[10] Pablo G Cámara. Topological methods for genomics: present and future directions. *Current Opinion in Systems Biology*, 1:95–101, 2017.

[11] Charisse Petersen and June L Round. Defining dysbiosis and its influence on host immunity and disease. *Cellular Microbiology*, 19(4):e12726, 2017.

[12] Ruth E Ley, Peter J Turnbaugh, Samuel Klein, and Jeffrey I Gordon. Microbial ecology: human gut microbes associated with obesity. *Nature*, 444(7122):1022–1023, 2006.

[13] Daniel McDonald, Embriette Hyde, Justine W Debelius, James T Morton, Antonio Gonzalez, Gail Ackermann, Alexander A Aksenov, Bahar Behsaz, Caitriona Brennan, Yingfeng Chen, et al. American Gut: an open platform for citizen science microbiome research. *mSystems*, 3(3):e00031–18, 2018.

[14] John Aitchison. The statistical analysis of compositional data. *Journal of the Royal Statistical Society: Series B*, 44(2):139–160, 1982.

[15] Gregory B Gloor, Jean M Macklaim, Vera Pawlowsky-Glahn, and Juan José Egozcue. Microbiome datasets are compositional: and this is not optional. *Frontiers in Microbiology*, 8:2224, 2017.

[16] Christopher Tralie, Nathaniel Saul, and Rami Bar-On. Ripser.py: A lean persistent homology library for Python. *Journal of Open Source Software*, 3(29):925, 2018.